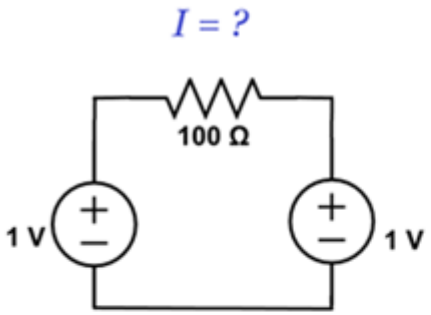
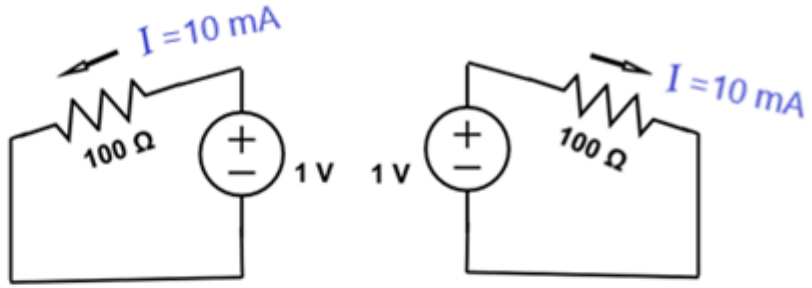
EE 312

Exp 2

PN JUNCTION DIODES AND THEIR CHARACTERISTICS

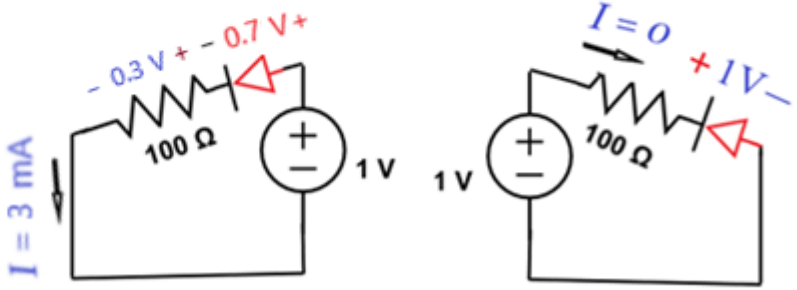
Diode vs. Resistor (Revision)





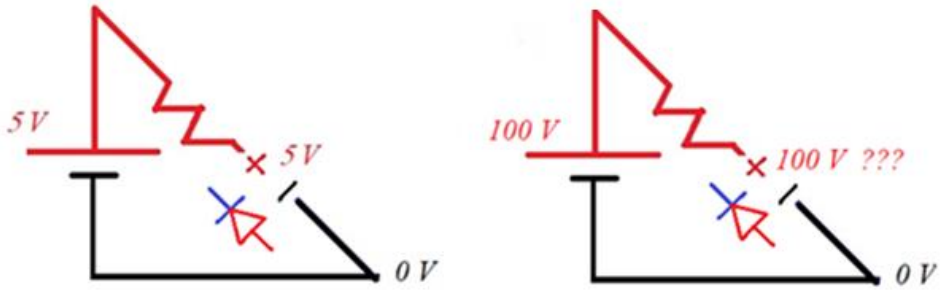
Without a Diode

Current can only flow in a resistor if there IS a potential difference. It flows from the higher potential terminal to the lower potentail terminal.

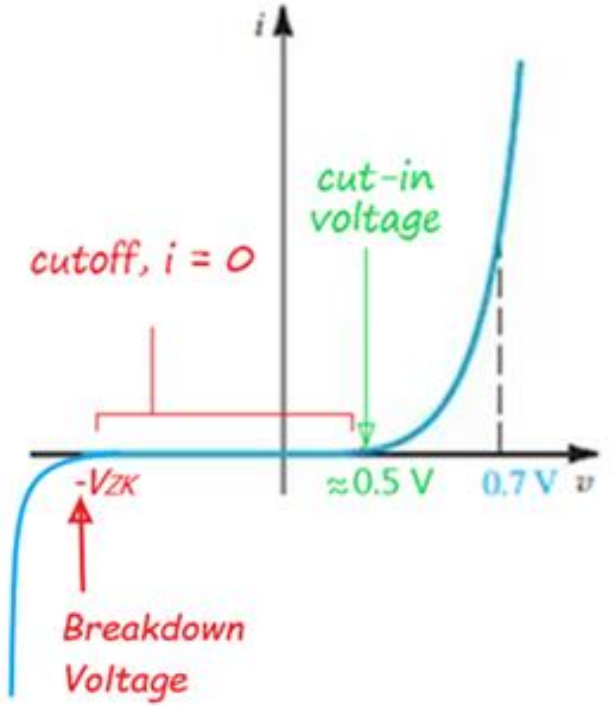


With a Diode!

Real Diodes Can Not Withstand Unlimited Reverse Biasing!



The supply voltage must be < V_{ZK}; otherwise breakdown will occur!



IN THIS EXPERIMENT:

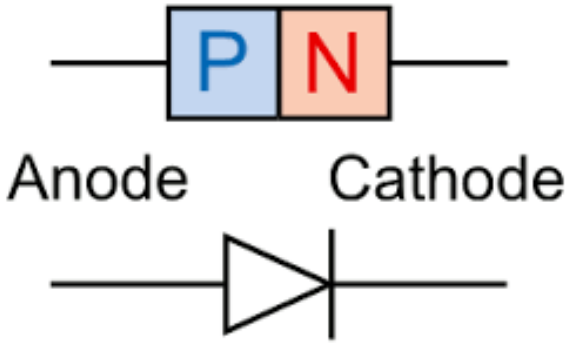
- Obtain the *i-v* characteristic curves for a rectifier diode and a Zener diode.
- Calculate Some Important Diode Parameters.
- Extract a piecewise-linear model for the Zener diode from its *i-v* characteristic curve.

JUNCTION DIODE

- Holes are vacancies in chemical bonds that can conduct current in the direction of an electric field. Movement of holes is equivalent to movement of valence electrons in the opposite direction.
- A P-type Semiconductor has:

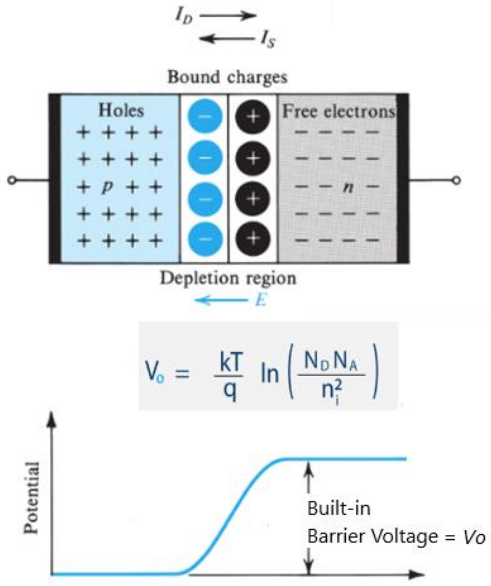
  Concentration of Holes >> Concentration of Electrons
- An n-type Semiconductor has:

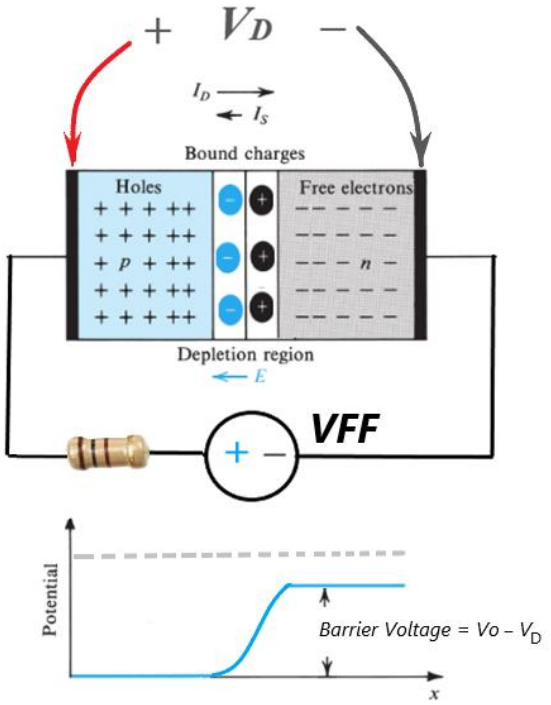
  Concentration of Electrons >> Concentration of Holes
- A Junction Between a P-type Semiconductor and another of N-type will form A Diode!



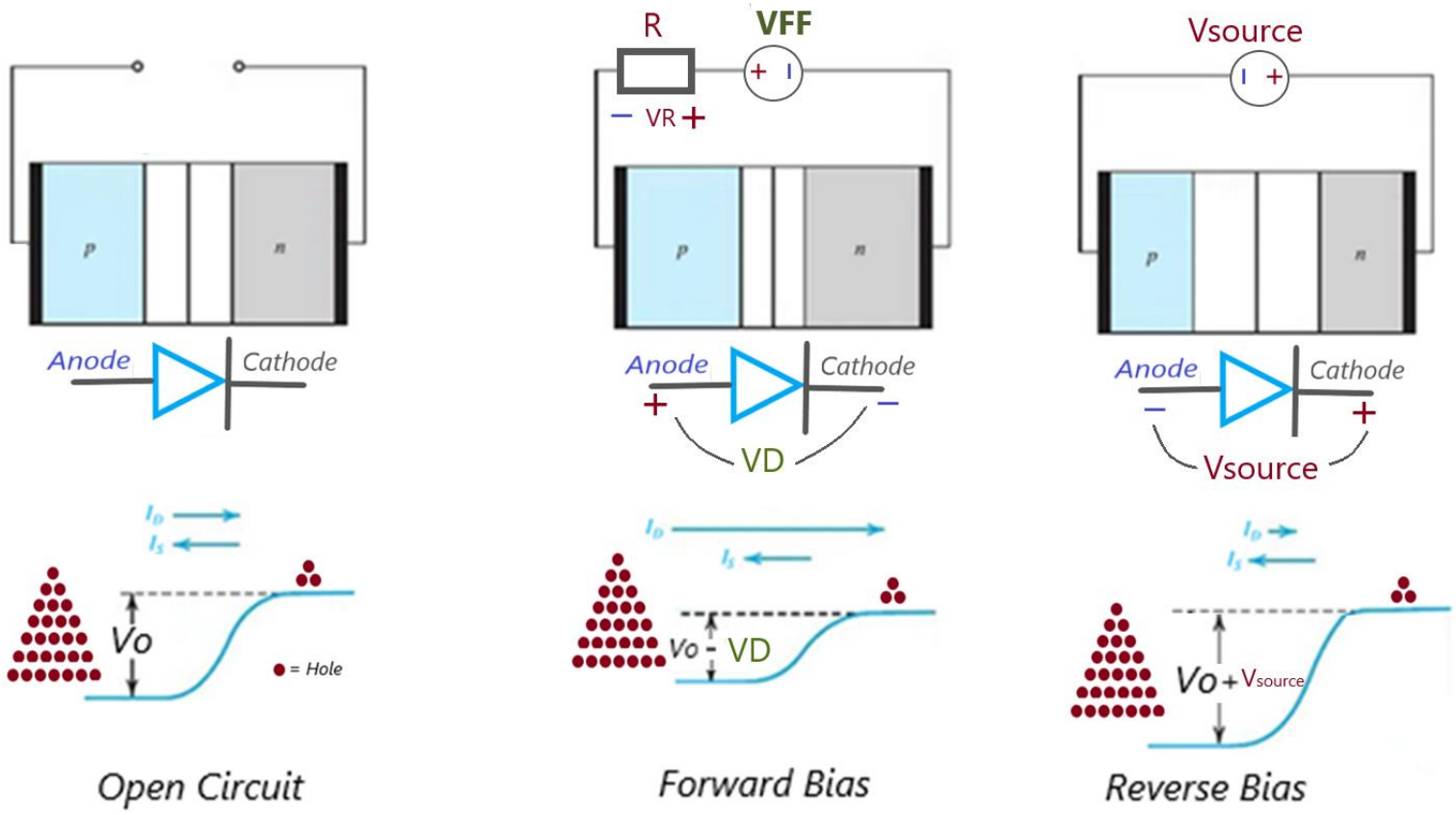
Built-In Barrier Voltage Vo

From the p-side of an Open-Circuit Junction Diode, there is a Built-In Potential Energy Barrier for Holes. This Barrier Voltage Can be Reduced by Forward-Biasing the Junction.





Forward Bias vs. Reverse Bias



THE DIODE *i-v* CHARACTERISTIC CURVE

$$i = I_S\left(e^{\frac{v}{nV_T}} - 1\right)$$

where I_S is the **saturation current**,
V_T is the thermal voltage ≅ 26 mV at $T = 300$ K,
n is the emission coefficient whose value $1 \le n \le 2$.

$$n = \frac{v_2 - v_1}{2.3 \times V_T \times \log\left(\frac{i_2}{i_1}\right)}$$

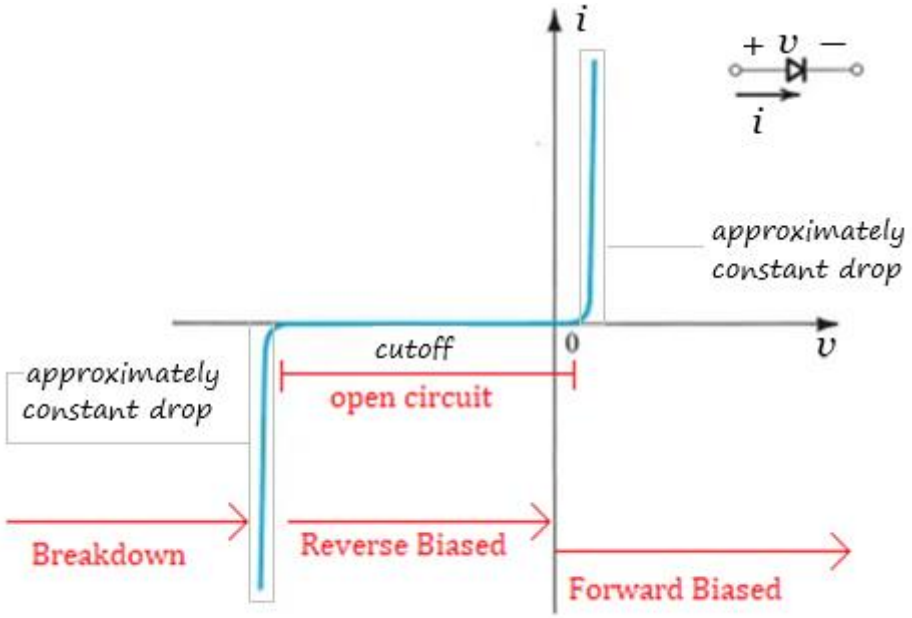
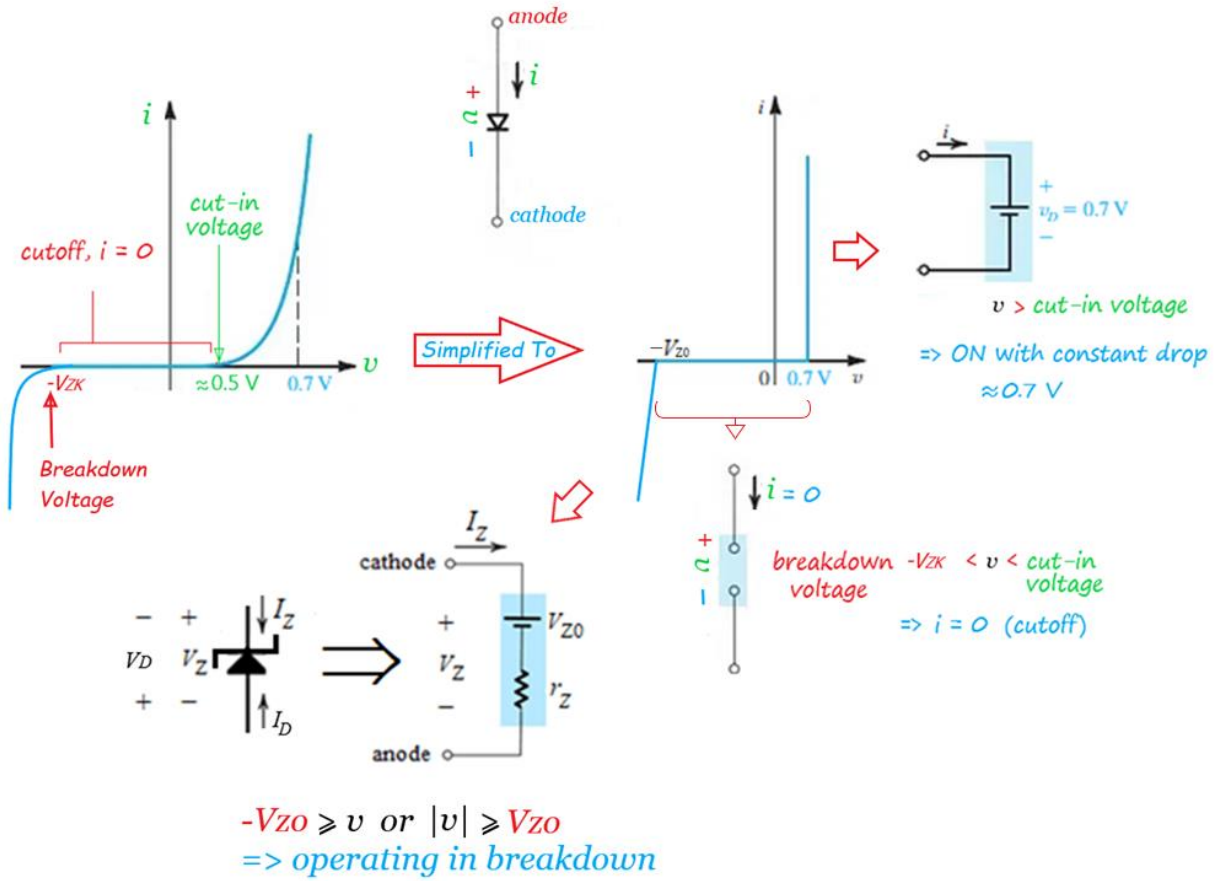


Figure 4.7 The i–v characteristic of a silicon junction diode.

Simplified Approximate Models for the Silicon Diode



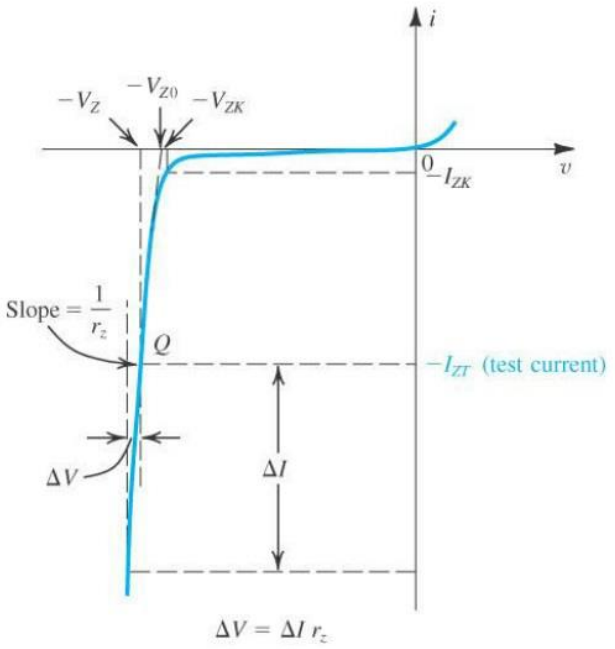
Is Reverse Breakdown Safe?

Breakdown is Considered to be Safe if the Resulting Reverse Current is Limited by enough resistance in the Circuit.

Zener Diodes are Safely Used in the Breakdown region to Regulate Voltage.

Rectifier Diodes used in (ac to dc) Conversion Must not be Subjected to a Reverse Voltage Greater in Magnitude than its Breakdown Voltage!

PIECE-WISE LINEAR MODEL in the BREAKDOWN REGION of a ZENER DIODE



$$r_Z = \frac{1}{Slope} = \frac{V_{Z2} - V_{Z1}}{I_{Z2} - I_{Z1}}$$

r_Z is denoted Z_{ZT} in **datasheets**

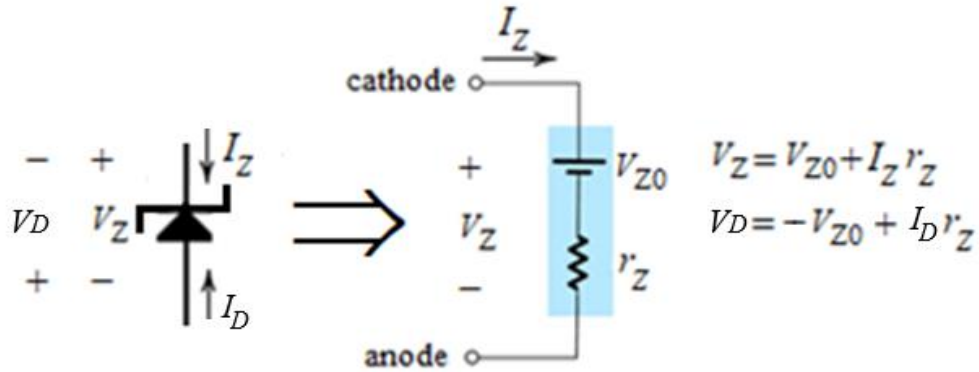


Figure 3.21 The diode *i–v* characteristic with the breakdown region shown in some detail.

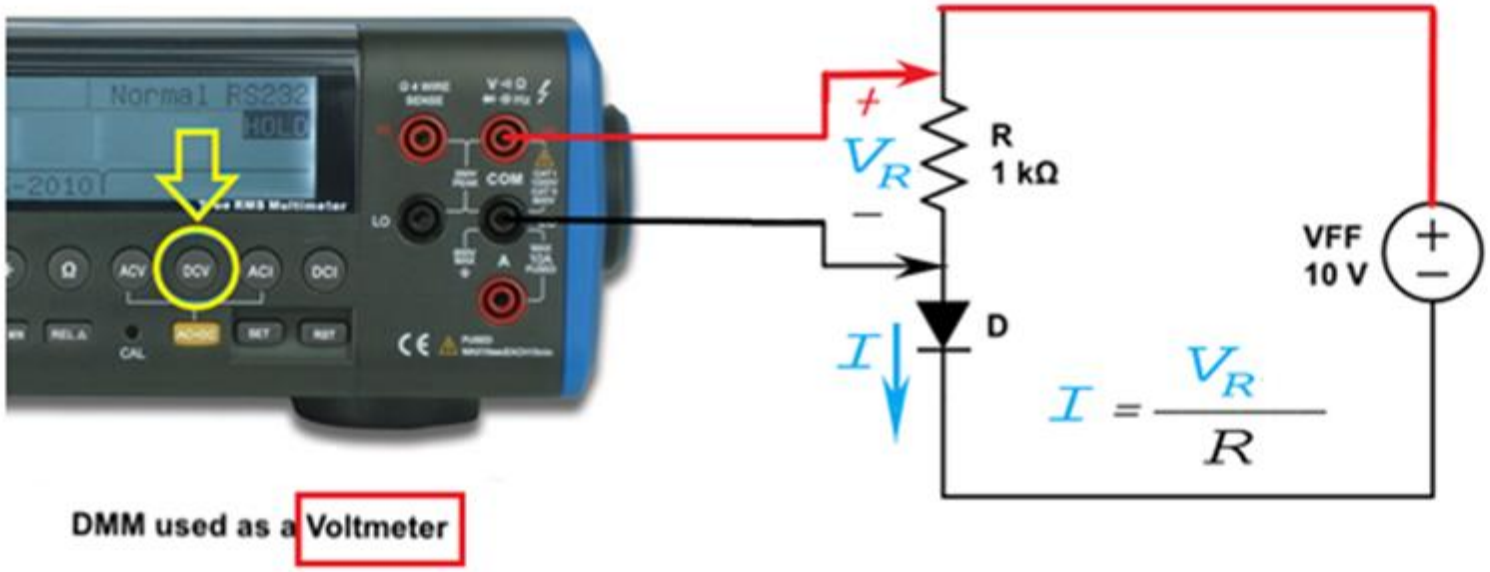
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EXPERIMENTAL PART

ROUNDING NUMBERS

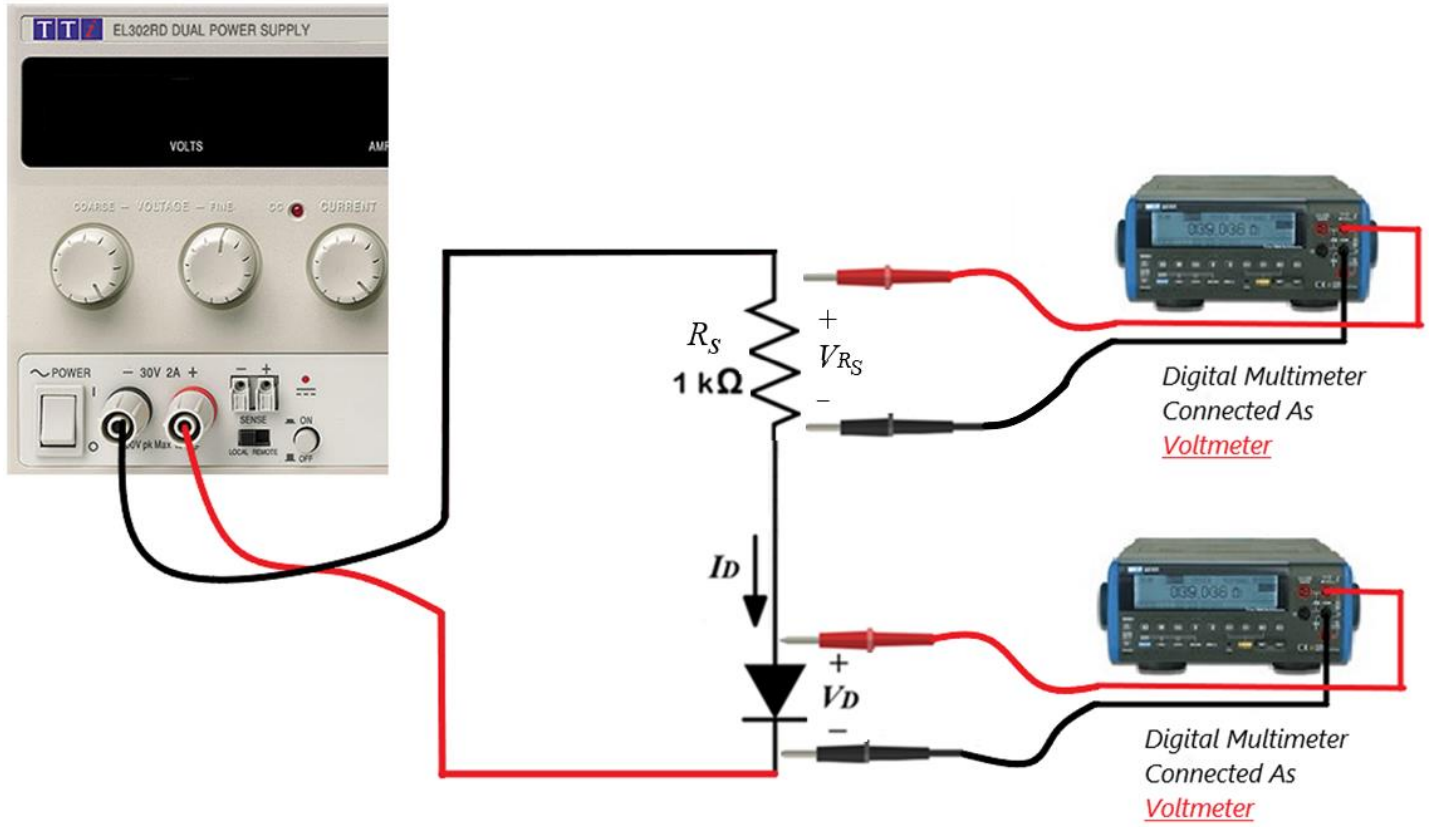


Measuring the Current Indirectly

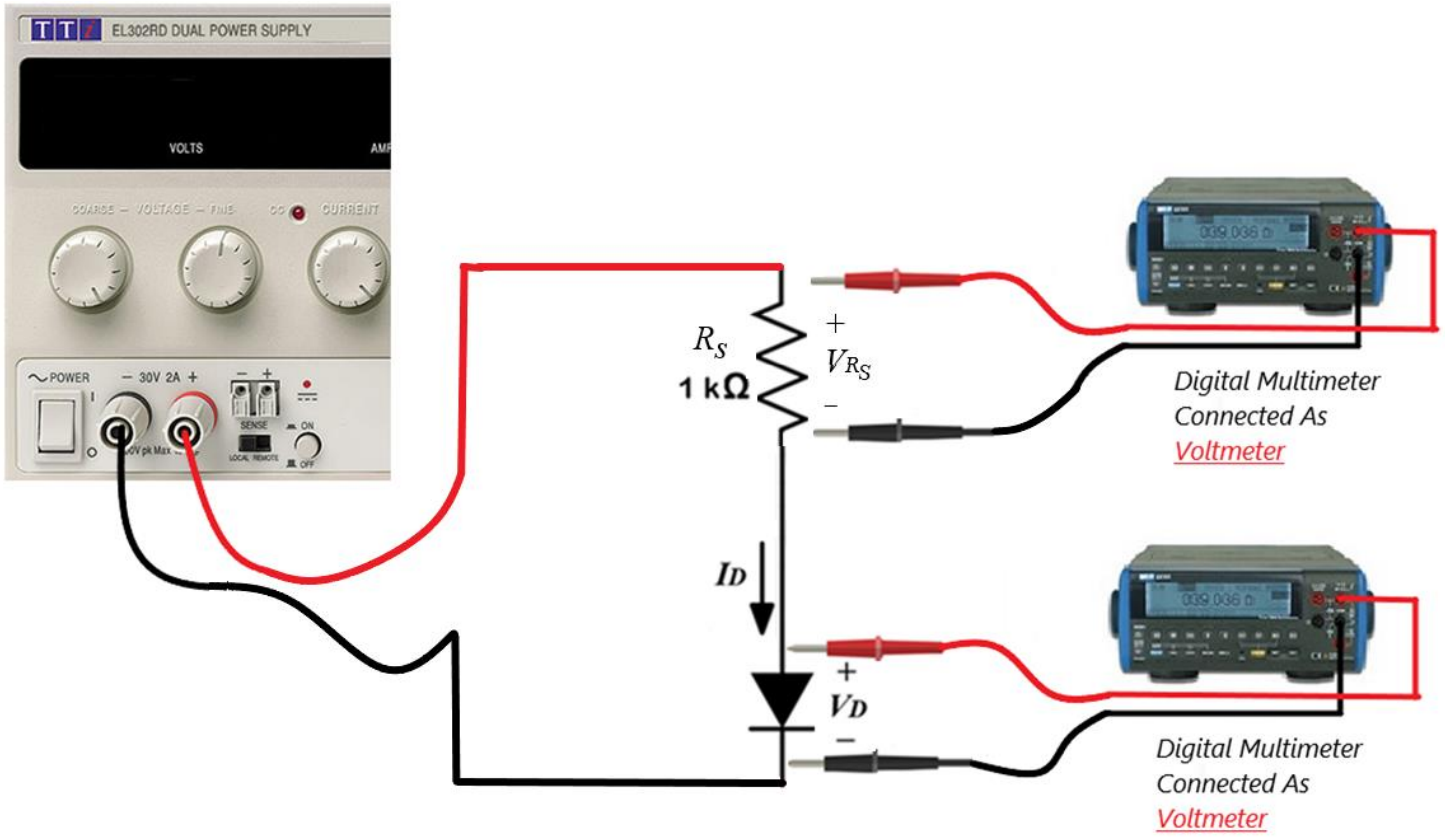


DMM used as a Voltmeter

Reverse-Bias Setup



Forward-Bias Setup



References

- A. S. Sedra and K. C.Smith, *Microelectronic Circuits*, New York: Oxford University Press, 2010.
- T. A. Almadhi, EE 312 Lab Manual:https://lms.ksu.edu.sa/bbcswebdav/users/talmadhi/EE312/EE312.pdf